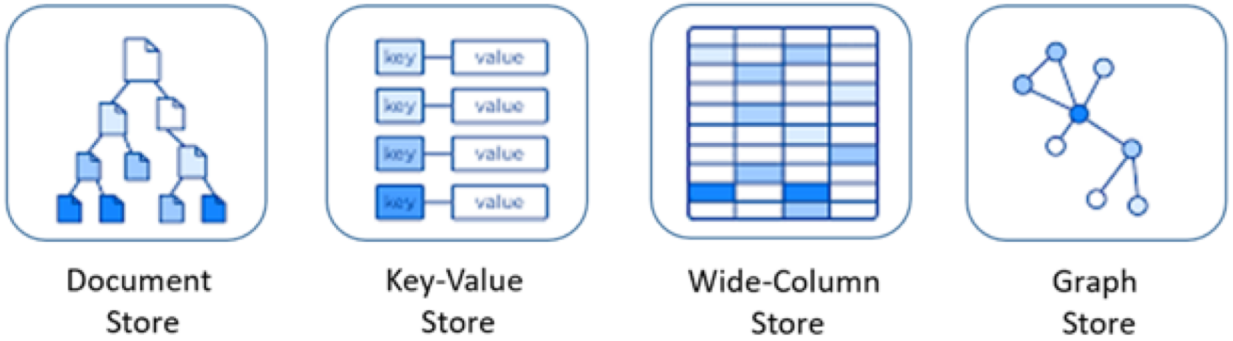
Wide Column Stores (WCS)

Wide column stores (a.k.a. extensible record stores) store data in records with an ability to hold very large numbers of dynamic columns (billions). Since the column names as well as the record keys are not fixed, wide column stores can be seen as two-dimensional key-value stores.

Google's **BigTable** is considered to be the origin of this class of databases. Classical work: "Bigtable: A Distributed Storage System for Structured Data" (2006) by Fay Chang et al - http://static.googleusercontent.com/media/research.google.com/en//archive/bigtable-osdi06.pdf

Popular examples of WCS databases:

- **Cassandra** (WCS based on ideas of BigTable and DynamoDB)
- **HBase** (WCS on top of Hadoop)
- **Microsoft Azure Cosmos DB** (Document store , Graph DBMS, Key-value store, WCS



Key	Product	Line	Size	Color	Color 2	Genera	Screen Size	Picture	Backlight	OS	Price	Qty	Cost
1	Shoes	Shoes	12								$50	1	$50
2	Jacket	Clothes	L	Blue	Gray						$69	1	$69
3	Apple Watch 3	Electronics	Large	Gold		3					$699	1	$699
4	Samsung SmartTV	Electronics				8	65"	4K	LED	Apple watchOS 4.0	$797	1	$797